MAHENDRA YADAV

Senior Software Engineer | Distributed Systems | Cloud Platforms

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PROFESSIONAL SUMMARY

Senior Software Engineer with 9+ years designing, building, and operating cloud-native distributed systems and internal platform services in Go and Python, with hands-on depth across AWS, Azure, and Google Cloud (Google Cloud and Azure certified) and infrastructure as code using Terraform and Ansible, experienced in Kubernetes security hardening, event-driven architecture, CI/CD, and observability, and in building AI-integrated services and agentic workflows consumed by multiple engineering teams, with a track record of owning components from ambiguous problem to production, mentoring engineers, and raising engineering standards

TECHNICAL SKILLS

Languages: Go, Python, Node.js, Elixir

Cloud & Platform: AWS, Azure, Google Cloud, Kubernetes, Terraform, Ansible, Helm, Docker, Calico

CI/CD, Observability & Security: Jenkins, CloudBees, GitLab, Bamboo, JFrog, Nexus, SonarQube, Grafana, Jaeger, OpenTelemetry, Azure Monitor, Kubernetes network policies, Argo CD, PII protection, application security hardening

Distributed Systems & APIs: Microservices, Event-Driven Architecture, gRPC, REST, Protocol Buffers, Kafka, Debezium

AI & Applied ML: LangChain, LangGraph, LangSmith, Snowflake Cortex Analyst, Snowflake Cortex Agent, LLM Prompting, LLM Evaluation, MLflow, CD4ML

Architecture & Practices: System Design, Distributed Systems, Infrastructure as Code, TDD, Data Cataloging (Purview)

Data & Databases: PostgreSQL, MySQL, Redis, Snowflake, Databricks (Spark), HDFS, Airflow

Frontend & Frameworks: React, Django, Flask, FastAPI, Gin, Phoenix

CERTIFICATIONS

Google Cloud Professional Cloud Architect | Google Cloud Associate Cloud Engineer | Microsoft Azure Administrator Associate (AZ-104) | Microsoft Azure Developer Associate (AZ-204)

PROFESSIONAL EXPERIENCE

Senior Software Engineer | ServiceNow, Hyderabad, Telangana, India

October 2024 - Present

- Led a 6-member engineering team on competitive-intelligence initiatives across unstructured data sources, reducing manual research time by ~30% and expanding coverage to 5 new data domains
- Architected and built an internal platform API around Snowflake Cortex Analyst and Agent, designing it as a shared, governed service consumed by ~8 internal engineering teams and handling ~50,000 requests/day
- Drove cross-functional integration of the platform API with enterprise AI and agentic workflows and Microsoft Teams, applying API design, security, scalability, and operational controls to production integrations
- Reviewed code and raised engineering standards across the team decomposed EPICs into deliverable work, improving sprint predictability by ~20%
- Owned infrastructure roadmap execution for a big-data platform, restructuring legacy components to reduce operational overhead by ~10 hours/week
- Designed and built an alerting service from scratch to improve operational visibility, reducing incident detection time from ~45 minutes to under 5 minutes
- Mentored 3 engineers, supporting their growth into senior individual-contributor roles

Application Developer (Lead Consultant) | Thoughtworks, Hyderabad, Telangana, India

March 2020 - October 2024

Project: Health Promotion Board, Singapore

- Owned infrastructure architecture and execution for a national health-data platform serving ~500,000 users
- Built Go-based internal platform services for access management and reporting, adopted as standard tooling across the engineering organization
- Designed and implemented a hub-and-spoke Azure Databricks architecture, reducing cross-workspace compute costs by ~20%
- Led security evaluation and hardening of a data-catalog application, closing 12 critical findings
- Diagnosed and resolved critical stability issues in an HDInsight Spark cluster, reducing job failure rate by ~40%
- Configured centralized logging and alerting via Azure Monitor, reducing mean time to detection by ~50%
- Proposed and executed cloud cost-optimization measures, saving ~$8,000/month in Azure spend
- Operated an HDInsight Kafka cluster with Debezium for event-driven, real-time CDC streaming, processing ~2 million events/day
- Hardened Kubernetes cluster security using Calico network policies, reducing the pod-to-pod attack surface

- Built large-scale batch processing pipelines using Python, Airflow, HDInsight (Kafka), and Databricks (Spark), processing ~5 TB of data daily
- Mentored new joiners and colleagues on infrastructure and data-engineering best practices

Project: Daimler - Data Ops

- Built CI/CD pipelines for data and ML projects, cutting model deployment time from ~3 days to a few hours
- Implemented CD4ML practices to standardize training and deployment of ML models across data science teams
- Used MLflow, DVC, Jenkins, and HDFS for versioned, large-scale data storage and experiment tracking

Project: Telemedicine Platform

- Partnered with the team to secure PII data across the platform DevOps pipeline, supporting healthcare data-protection compliance
- Used AWS Data Migration Service to provision non-PII data for consumer-facing access, unblocking downstream analytics use cases

Full Stack Developer | BeautifulCode, Hyderabad, Telangana, India

March 2018 - March 2020

Project: Campaign Management Platform, Sojern (Travel Ad-Tech)

- Built microservices for a campaign-management platform using Python, Django, gRPC, and React, supporting ~150 active campaigns/day
- Developed dynamic infrastructure-provisioning scripts on GCP, enabling self-service test environments and cutting provisioning turnaround from ~4 hours to ~15 minutes
- Owned infrastructure provisioning and operations for development and test environments
- Designed and implemented a scalable Go and gRPC notification service handling ~100,000 notifications/day across campaign and booking status alerts

Jr Software Engineer | Azri Solutions, Hyderabad, Telangana, India

December 2016 - March 2018

Project: GRNConnect

- Automated data ingestion and ETL processes using Python and Apache Airflow, eliminating ~15 hours/week of manual effort
- Built Elixir/Phoenix microservices for a hospitality-industry client, supporting ~500 concurrent users

EDUCATION & LEADERSHIP

Bachelor of Technology in Computer Science & Engineering, Dr B C Roy Engineering College, Durgapur, West Bengal | 2013 - 2017

- Organized internal technology-community knowledge-sharing sessions, growing regular attendance to ~30 engineers
- Mentored 5 new joiners, supporting onboarding and technical ramp-up
- Presented talks internally and at local communities on Docker, containerization, and database sharding
- Presented "Managing Data Pipelines using Airflow" at PyConf Hyderabad 2017